A Multimodal System for Automated Corn Disease Detection and Farmer-Centric Knowledge Sharing

25-26J-239

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology specialized in Information Technology

Department of Information Technology

Faculty of Computing

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

This research proposes the creation and evaluation of an integrated data-driven system that leverages state-of-the-art machine learning, computer vision, and explainable AI techniques to enhance corn productivity and resilience in smallholder production systems. The research addresses the most acute challenges facing corn farmers, including early disease detection, pest control, precise nutrient application, and yield estimation under diverse environmental conditions. The general objective is to provide farmers with timely, accurate, and interpretable data that may improve decision-making as well as minimize wastage of resources.

As part of the strategy, the project will develop robust computer vision models for the detection of corn diseases using leaf images. A tested dataset of corn leaf images will be collected and preprocessed to maintain diversity and good quality. Various deep learning models will be trained and compared systematically to establish the most appropriate methods for disease detection. Explainable AI methods will be integrated to provide transparency and trustworthiness in such a way that farmers and agribusiness stakeholders can understand the justification of predictions provided by the models. Optimization techniques will be employed to adapt the models to low-resource devices to ensure accessibility and usability in smallholder farming contexts where computational resources are limited.

Through this approach, the proposed research seeks to introduce a scalable and interpretable solution that will be capable of empowering corn farmers with meaningful information, ultimately fostering sustainable agricultural management as well as improving crop tolerance to biotic stressors.

Introduction

Corn (Zea mays L.) underpins food security, feed, and income for millions of smallholders, yet yields are persistently threatened by foliar diseases whose incidence and geographic suitability are projected to expand under climate change, amplifying losses if detection is delayed. Manual scouting remains the norm but is labor-intensive, subjective, and often inaccessible in resource-limited settings, motivating automated, data-driven diagnostics tailored to smallholder contexts.

Over the last decade, **computer vision and deep learning** have shown strong promise for plant disease recognition from leaf images, with numerous studies reporting high test accuracy on benchmark datasets such as PlantVillage. However, generalization from controlled datasets to field conditions can be limited by background bias, lighting, occlusion, and cultivar variability; further, many high-performing models are computationally heavy for edge/mobile deployment.

To improve trust and usability, the community increasingly incorporates **explainable AI (XAI)**—e.g., Grad-CAM—to highlight image regions that drive predictions, enabling agronomists and farmers to verify if models attend to pathognomonic symptoms rather than spurious cues. Parallel work explores lightweight or efficient backbones (e.g., MobileNet/EfficientNet) and bespoke compact architectures for **on-device inference**, balancing accuracy with memory/FLOPs for real-time use. Recent maize-focused studies also test hybrid CNN–Transformer pipelines and field-validated models, indicating a broader state of the art that spans accuracy, interpretability, and deployability.

State of the art. Contemporary research combines (i) high-accuracy deep models trained on curated and emerging in-field datasets; (ii) XAI to provide saliency/attribution maps; and (iii) **edge-ready** networks and toolchains to run on smartphones or embedded devices. Surveys from 2024–2025 consolidate these trends and emphasize the gap between lab benchmarks and realistic farm environments, especially for smallholders

Our approach and contribution (this component). We will (1) **collect and preprocess** a diverse set of corn leaf images with field-realistic variation; (2) **train and compare** multiple deep learning architectures (including efficient backbones) for disease recognition; (3) integrate **XAI (e.g., Grad-CAM/LIME)** for transparent predictions; and (4) **optimize for low-resource devices** to enable offline, on-farm use. By explicitly co-optimizing accuracy, interpretability, and deployability—and validating beyond controlled datasets—this work builds directly on prior art while addressing critical barriers to adoption in smallholder corn systems.

Background & Literature survey

Corn (Zea mays L.) stands as one of the most pivotal cereal crops globally, serving as a staple food, livestock feed, and a vital raw material in various industries. Its versatility and high yield potential make it a cornerstone of agricultural economies, particularly in developing regions where smallholder farmers are the primary producers. In these areas, corn cultivation is not

merely an economic activity but a critical component of food security and rural livelihoods. However, the productivity of corn is frequently compromised by various biotic stresses, notably foliar diseases, which can lead to significant yield losses if not promptly identified and managed.

Challenges Posed by Foliar Diseases

Foliar diseases in corn, such as Northern Leaf Blight, Gray Leaf Spot, and Common Rust, are prevalent across diverse geographical regions. These diseases are characterized by distinct symptoms, including lesions, chlorosis, and necrosis, which can severely impair photosynthesis and transpiration processes. The timely identification of these symptoms is crucial, as delayed detection often leads to the spread of the disease, exacerbating yield losses. Traditional methods of disease diagnosis, relying on manual inspection by trained agronomists, are not only time-consuming but also prone to human error and subjectivity. This underscores the need for automated, reliable, and scalable solutions to detect and manage these diseases effectively.

Evolution of Disease Detection Methods

Historically, plant disease detection was based on visual assessment by experts, a method that, while effective, is limited by the availability of skilled personnel and the time constraints inherent in field assessments. The advent of digital imaging and computer vision introduced new avenues for disease detection. Early approaches utilized image processing techniques to analyze leaf textures and patterns, aiming to identify disease symptoms. However, these methods often struggled with variability in lighting conditions, leaf orientations, and background clutter, leading to inconsistent results.

The integration of machine learning (ML) techniques marked a significant advancement in disease detection. Supervised learning algorithms, such as Support Vector Machines (SVMs) and Random Forests, were employed to classify images based on extracted features. While these models demonstrated improved accuracy, their performance was still influenced by the quality and representativeness of the training data. Moreover, the manual feature extraction process was labor-intensive and required domain expertise.

Rise of Deep Learning in Plant Disease Detection

The introduction of deep learning (DL), particularly Convolutional Neural Networks (CNNs), revolutionized plant disease detection. DL models, with their ability to automatically learn hierarchical features from raw image data, have shown remarkable success in various image classification tasks. In agriculture, CNNs have been applied to classify plant diseases with high accuracy, often exceeding 90%. These models can learn complex patterns in leaf images, enabling them to distinguish between healthy and diseased plants effectively.

Despite their success, the deployment of DL models in real-world agricultural settings presents several challenges. One significant issue is the generalization of models trained on curated

datasets to field conditions, where variations in lighting, background, and leaf appearance can affect model performance. Additionally, DL models are often computationally intensive, requiring substantial processing power and memory, which may not be available in resource-constrained environments typical of smallholder farms.

Explainable AI: Enhancing Model Transparency

To address the "black-box" nature of DL models, Explainable AI (XAI) techniques have been developed to provide insights into model decision-making processes. Methods such as Grad-CAM and Local Interpretable Model-agnostic Explanations (LIME) generate visual or textual explanations that highlight the regions of input images influencing model predictions. In the context of plant disease detection, XAI can help validate that models focus on relevant disease symptoms rather than irrelevant features, thereby increasing trust among users and facilitating model adoption in agricultural practices.

Deployment Challenges and Edge Computing Solutions

Deploying DL models in real-world agricultural settings necessitates consideration of hardware constraints. Many smallholder farmers rely on mobile devices with limited processing capabilities and storage. To overcome these limitations, researchers have explored lightweight DL architectures, such as MobileNet and EfficientNet, which are designed to perform well on resource-constrained devices. These models balance accuracy with computational efficiency, making them suitable for deployment in field conditions.

Edge computing further enhances the feasibility of deploying DL models in agriculture. By processing data locally on devices, edge computing reduces the reliance on cloud infrastructure, mitigating issues related to connectivity and data privacy. Edge AI systems can provide real-time disease detection, enabling timely interventions and reducing the need for expert involvement.

Current Trends and Future Directions

Recent advancements in DL, XAI, and edge computing have paved the way for integrated systems capable of real-time, transparent, and efficient plant disease detection. However, challenges remain in developing models that generalize well across diverse field conditions, ensuring interpretability for end-users, and optimizing models for deployment on low-resource devices. Ongoing research focuses on addressing these issues, aiming to create scalable solutions that empower smallholder farmers to manage plant diseases effectively and sustainably.

Research Gap

Several studies have addressed automated plant disease detection using machine learning and deep learning techniques. While these studies show promising results, they often focus on specific aspects such as accuracy, dataset evaluation, or model architecture, without fully integrating explainability, field deployment, or usability for smallholder farmers. Among the most relevant studies are **Research A**, **Research B**, and **Research C**, summarized in Table 1.1.

Table 1.1: Research Gap Comparison

Application / Feature References	Research A	Research B	Research C	Proposed System
Disease detection from leaf images	✓	✓	✓	✓
Use of deep learning models	✓	✓	✓	✓
Explainable AI / Transparent predictions	✗	✓	✗	✓
Edge / low-resource deployment	✗	✗	✗	✓
Field-realistic dataset	✓	✗	✓	✓
Targeted for smallholder farmers	✗	✗	✗	✓

Research A focuses on corn disease detection using CNN-based models trained on the PlantVillage dataset. This research demonstrates high classification accuracy under controlled conditions but does not include explainable AI features or model deployment on mobile devices. Compared with the proposed system, Research A addresses only disease detection and the use of deep learning models.

Research B applies lightweight CNN architectures and includes some explainability methods for interpreting predictions. However, it primarily uses curated datasets and does not test deployment in real field environments or optimize models for smallholder contexts. Compared with the proposed system, Research B covers explainable AI but lacks full deployment feasibility and field-ready datasets.

Research C investigates edge deployment of lightweight plant disease detection models, showing potential for mobile applications. Nevertheless, the study does not integrate explainable AI and focuses mostly on model efficiency rather than transparency or usability for farmers.

Compared with the proposed system, Research C addresses edge deployment and field feasibility but lacks explainable AI and smallholder-specific design considerations.

The proposed system builds upon these studies by combining **accurate disease detection, deep learning, explainable AI, and edge deployment**, all tailored for smallholder farmers using field-realistic datasets. This comprehensive approach addresses the gaps identified in previous research, ensuring practical usability, interpretability, and accessibility in real-world agricultural settings.

Research problem

Corn leaf diseases such as Northern Leaf Blight, Gray Leaf Spot, and Common Rust cause severe yield losses worldwide, and early detection is critical for preventing large-scale damage. Previous research has explored the use of deep learning models like ResNet, VGG16, YOLO, and FourCropNet for automated disease detection. While these models perform well on controlled datasets such as PlantVillage, they face several limitations when applied in real-world farming environments.

Most datasets are collected under controlled conditions with uniform lighting, plain backgrounds, and clear images of individual leaves. In contrast, farm environments include variable lighting, shadows, overlapping leaves, dust, and background noise such as soil or other plants, which significantly reduce detection accuracy. Additionally, these models often require high computational resources, making them impractical for deployment on low-resource devices like smartphones that farmers typically use. Many existing systems are also designed from a technical perspective without considering usability or farmer adoption, meaning predictions are not transparent, easy to interpret, or actionable.

As a result, current solutions, though promising in accuracy under lab conditions, lack field adaptability, resource efficiency, and farmer-centric usability, creating a gap between research outputs and practical solutions that farmers in regions like Sri Lanka can benefit from.

Evaluation

System evaluation will be conducted to assess technical performance, usability, and deployment feasibility:

- Technical Performance:

  Model accuracy, precision, recall, and F1-score will be calculated to quantify the reliability of pest detection. The reduction of false positives and negatives will also be assessed.

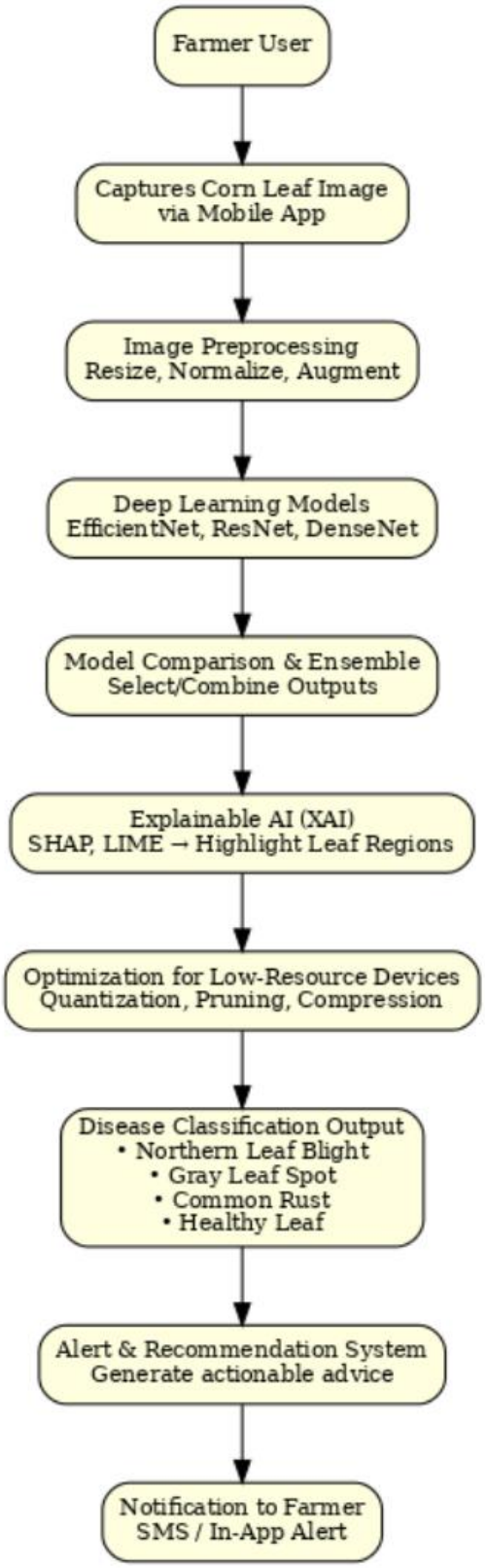
- Usability Assessment:

  Farmers will participate in usability tests to provide feedback on alert effectiveness, ease of interpretation, and overall system satisfaction.

- Deployment Feasibility:

  The system's ability to function in low-resource settings, including limited internet connectivity and low-end devices, will be tested. Scalability and adaptability to diverse farm environments will also be considered.

Conceptual frameworks



Non-Functional Requirements

The non-functional requirements ensure the system operates effectively, efficiently, and reliably in real-world environments:

- ### Performance

  The system should deliver detection results and alerts within seconds, enabling timely intervention to prevent crop damage. High computational efficiency is critical to meet the requirements of real-time monitoring.

- ### Scalability

  The system must support a large number of simultaneous users, image uploads, and alert notifications without degradation in performance. Scalability ensures the system can grow to accommodate wider adoption by farming communities.

- ### Reliability

  Detection models must minimize false positives and negatives, achieving an accuracy of at least 90%. The system should maintain consistent performance even under high data loads or varying environmental conditions.

- ### Security

  Farmer data, images, and system logs must be securely stored. Privacy protection measures, including secure transmission and storage protocols, must be implemented to prevent unauthorized access and ensure compliance with data protection standards.

- ### Usability

  The interface should be intuitive and user-friendly, accommodating farmers with varying levels of digital literacy. Clear navigation, simple input forms, and accessible alert presentation are key usability considerations.

- ### Availability

  The system should maintain near-continuous operation (24/7 uptime) with minimal downtime to ensure farmers can rely on timely alerts and access system features whenever needed.